Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - II

Course Code: BCA104-2 **Max Marks: 100**

Course Name: DISCRETE MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate a working knowledge of set notation and elementary set theory and recognize the connection between set operations and logic.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effective express the elementary concepts and results of Number theory.

Section A **3*8=24**

1)	Explain how to determine if a given relation on a set is antisymmetric using an example and describe its connection to logical implications.	**3**	(CO1, L2)
2) a)	Discuss how a function can be classified as an injection, surjection, or bijection with examples from different sets. Provide a small calculation to illustrate your explanation.	**3**	(CO2, L2)
	OR		
b)	Describe an algorithm to find the modular inverse of an integer 'a' modulo 'm', and use it to find the modular inverse of 7 modulo 25.	**3**	(CO2, L2)
3)	Describe an algorithm to determine if a given square matrix is symmetric and provide a small example.	**3**	(CO3, L2)
4)	Determine if the relation R = {(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3)} on the set A = {1, 2, 3} is symmetric. Justify your answer.	**3**	(CO4, L3)
5) a)	Determine if the function : $\mathbb{N} \rightarrow \mathbb{Z}$ defined by () = 2 + 1 is injective. If it is, prove your claim; otherwise, provide a counterexample.	**3**	(CO5, L3)
	OR		
b)	Demonstrate that the function : $\mathbb{N} \rightarrow \mathbb{N}$ defined by () = 3 + 2 is injective but not surjective. Show your reasoning and provide an example to support your answer.	**3**	(CO5, L3)
6)	Determine if the given relation R = {(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)} on the set A = {1, 2, 3} is symmetric and explain your answer.	**3**	(CO1, L3)

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7) a)	Examine and differentiate between an injective function and a surjective function with examples. Provide a small calculation or explanation for each example.	**3**	(CO2, L4)
	OR		
b)	Examine and categorize the given functions based on their properties as injections, surjections, or bijections: f: R → R defined by f(x) = 2x + 3 and g: Z+ → Z+ (set of positive integers) defined by g(n) = n^2. Justify your answers.	**3**	(CO2, L4)
8)	Construct a matrix representation for the symmetric relation R on the set A = {1, 2, 3, 4} given by R = {(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1)}. Also, determine if this relation is reflexive and explain your reasoning.	**3**	(CO3, L3)
	Section B		**8*7=56**
9)	Consider the relation defined on the set = {1, 2, 3, 4} by the ordered pairs: = {(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (3,4)}. Determine and prove whether the relation is reflexive, symmetric, antisymmetric, or transitive. If not, explain why it fails to satisfy each property. Additionally, construct an equivalence relation on that contains as a subset of its ordered pairs, and demonstrate how this new relation satisfies all four properties.	**8**	(CO4, L3)
10) a)	Analyze the Euclidean algorithm for finding the greatest common divisor (GCD) of two integers. Using this algorithm, determine the GCD of [546, 714]. Additionally, compare and contrast the Euclidean algorithm with another method to find the GCD, explaining the advantages and disadvantages of each.	**8**	(CO5, L4)
	OR		
b)	Examine and prove which of the following pairs of functions defined over the sets A = {1, 2, 3, 4} are bijective (one-to-one and onto). For each function that is bijective, find its inverse and verify your result by showing that f(f^{-1}(x)) = x and f^{-1}(f(x)) = x for all x in the domain. Additionally, analyze whether these functions form a symmetric relation with respect to their elements in A. Provide detailed proofs for each part of your analysis.	**8**	(CO5, L4)
11)	Discuss the concept of countability in sets and prove that the set of all rational numbers between 0 and 1 is countable. Additionally, explain why the set of natural numbers is also countable.	**8**	(CO1, L2)
12) a)	Given two sets A = {1, 2, 3, 4} and B = {a, b, c}, let R1 be a relation from A to B defined by R1 = {(1, a), (2, b), (3, c), (4, a)}, and let R2 be a relation from B to A defined by R2 = {(a, 1), (b, 2), (c, 3)}. Demonstrate the following operations on these relations: (i) Calculate the composition of functions R1 o R2. (ii) Determine if R1 is injective, surjective, or bijective. (iii) Find the inverse relation R1^-1 and verify that it satisfies the properties of an inverse function.	**8**	(CO2, L3)
	OR		
b)	Design an algorithm to find integers a and b such that they satisfy the equation 17a + 23b = gcd(17, 23), where gcd denotes the greatest common divisor. Demonstrate your algorithm by finding specific values of a and b.	**8**	(CO2, L3)
13)	Analyze two matrices representing relations on sets A and B as follows: Matrix M1 = [0 1 0; 1 0 1; 0 1 0] and matrix M2 = [1 0 0; 0 1 0; 1 0 0], where the rows represent set A and columns represent set B. Determine (i) whether each relation represented by M1 and M2 is symmetric, reflexive, or transitive individually; (ii) find the matrix that represents the union of these relations; (iii) calculate the composition of the two relations using matrix multiplication.	**8**	(CO3, L4)

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14) a)	Explain with examples how to determine if a given binary relation on a set is reflexive, symmetric, antisymmetric, and transitive. Also, describe the properties of equivalence relations and partial orderings. Illustrate your answer by considering the following sets and relations (i) = {1, 2, 3} and the relation = {(1,1), (2,2), (3,3), (1,2), (2,1)}; (ii) = {a, b, c} and the relation = {(a,a), (b,b), (c,c), (a,b), (b,c)}.	**8**	(CO4, L2)
	OR		
b)	Consider the relation R on the set A = {a, b, c, d} defined by R = {(a, a), (b, b), (c, c), (d, d), (a, b), (b, c), (c, d)}. Discuss whether R is reflexive, symmetric, antisymmetric, and transitive. If not, modify the relation to make it an equivalence relation or a partial ordering, providing justification for your modifications.	**8**	(CO4, L2)
15)	Analyze the Euclidean algorithm to find the greatest common divisor (GCD) of two integers. Using your analysis, determine the GCD of [390, 154] and demonstrate each step of the algorithm. Additionally, explain how the concept of the GCD can be applied to solve the problem of finding a linear combination of these two numbers that equals their GCD.	**8**	(CO5, L4)
	Section C		**10*2=20**
16)	Explain with examples how to determine if a given binary relation on a set is symmetric and provide an interpretation of this property in terms of directed graphs. Additionally, illustrate your explanation by considering two specific relations: 1 = {(1, 2), (2, 1), (3, 4)} and 2 = {(1, 2), (2, 3)}.	**10**	(CO1, L2)
17) a)	Consider two sets, A = {a, b, c} and B = {1, 2}. Let R_1 be a relation from A to B defined as R_1 = {(a, 1), (b, 2), (c, 1)} and let S be the identity relation on set A, i.e., S = {(a, a), (b, b), (c, c)}. Analyze the following properties of these relations and justify your answers using appropriate examples from the sets A and B: (i) Determine if R_1 is injective, surjective, or bijective. Justify your answer. (ii) Find the composition S R_1 and discuss its characteristics as a relation on set A. Is it reflexive, symmetric, or transitive? Explain why. (iii) Define an inverse function for R_1 if it exists; otherwise, explain why no such function can be defined.	**10**	(CO2, L4)
	OR		
b)	Examine and prove whether the union of two countable sets, and , forms another countable set. Additionally, differentiate between injections, surjections, and bijections in this context.	**10**	(CO2, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create